1. Dataset Exploration & Empirical Statistics

Distribution Shift

Task & Dataset Selection:

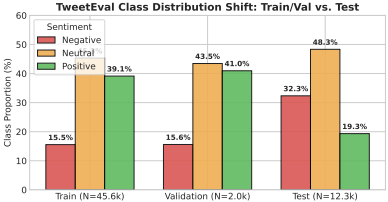
- **Dataset**: `cardiffnlp/tweet_eval` (Sentiment benchmark).
- **Split**: 45,615 Train, 2,000 Val, 12,284 Test ($K = 3$ classes).

Critical Empirical Finding: Severe Prior Shift:

- **Train**: 15.5% Neg, 45.3% Neu, 39.1% Pos.
- **Test**: **32.3% Neg** ($\times$**2.1**), 48.3% Neu, **19.3% Pos** ($\times$**0.5**).
- *Integrity*: Tempered class weighting $w_c \propto (1/N_c)^{0.5}$ and validation-calibrated prior multipliers prevent negative over-prediction without test leakage.

Polars Stylometric Insights:

- Median length: 19 words; high lexical noise and Twitter slang.
- **Negation cues**: 0.224/tweet in Neg vs. 0.079 in Pos (**3**$\times$ higher). Particles (*not, n't, never*) invert polarity.
- **Exclamations**: 0.468/tweet in Pos vs. 0.157 in Neg (**3**$\times$ higher).



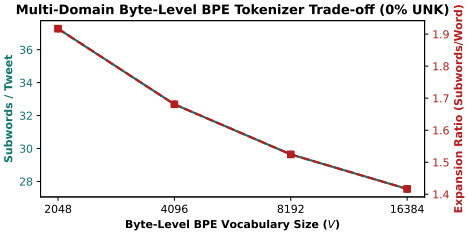
2. Text Segmentation & Tokenizer Design

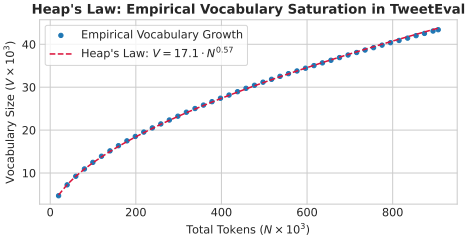
Why Byte-Level BPE over Whitespace Tokenization?

- **Heap's Law**: $V(N) = 17.14 \cdot N^{0.572}$. Whitespace tokenization yields $43,395$ raw types $\Rightarrow$ extreme sparsity and catastrophic test Out-Of-Vocabulary (OOV) rate.
- **Expanded Corpus Training**: Trained BPE on 63,650 multi-domain tweets (Sentiment + Emotion + Irony + Offensive).

Hyperparameter Sweep ($V \in \{2\mathbf{k}, 4\mathbf{k}, 8\mathbf{k}, 16\mathbf{k}\}$):

- Byte-level pre-tokenization maps any arbitrary byte $\Rightarrow$ **Strictly 0.00% character UNK rate**.
- **Optimal Size V $= 8,192$**: Yields 27.8 subwords/tweet (1.52 subwords/word). Preserves morphological negation affixes (*un-, dis-, -less, n't*) critical for sentiment polarity.



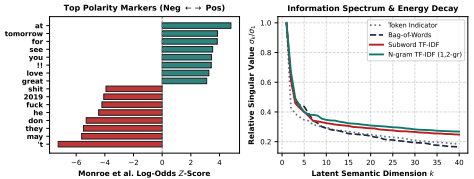


Five Representation Paradigms ($V = 8,192$):

1. **Token Indicator**: $x_i = \mathbb{I}(\text{token } i \in d) \in \{0, 1\}$. Binary presence; invariant to term repetition.
2. **Bag-of-Words (BoW)**: $x_i = \text{count}(i, d) \in \mathbb{N}_0$. Frequency counts of subword tokens.
3. **Subword TF-IDF**: L2-normalized with smoothed IDF.
4. **Subword N-Gram TF-IDF**: (1,2)-grams with sublinear scaling $1 + \log(\text{tf})$. Captures bigram collocations.
5. **Sequential Token IDs**: $x = [t_1, \ldots, t_{64}] \in \mathbb{N}^{64}$. Preserves word order, syntax, and negation scope.

Theoretical Note for Oral Defense:

- Monroe et al. Log-Odds identify salient sentiment markers.
- BoW destroys syntactic scope: *"not happy, very bad"* yields identical BoW to *"not bad, very happy"*.



Information Spectrum Insight

Singular value spectrum shows TF-IDF and N-gram spaces prevent high-frequency stopwords from dominating latent variance, enriching feature representation.

4. Model Architectures & Modern Training

Topic 4: Baselines vs. Deep MLP Ablation vs. Sequence Champions

Model Architectures Tested:

1. **Log-Linear Softmax**: $P(y|x) = \text{softmax}(Wx + b)$ (24.6k params).
2. **Shallow MLP**: $\text{ReLU}(W_1 x + b_1)$ with Dropout (2.10M params).
3. **Deep Residual MLP**: 4 layers, LayerNorm, GELU, skip connections (2.23M params). *Ablation: fails on BoW!*
4. **BiLSTM + Attention**: In-domain PPMI-SVD subwords, Conv1D stem, BiLSTM, Tri-Pooling + R-Drop (1.97M params).
5. **Transformer Scratch**: 2-layer Pre-LN Multi-Head Self-Attention + Positional Embeddings + Tri-Pooling (1.38M params).

Self-Supervised In-Domain Initialization (Zero External Data):

- Shifted PPMI + Truncated SVD (Levy & Goldberg, 2014) on 63.6k training tweets bootstraps subword semantic space.
- R-Drop symmetric KL consistency regularizes attention heads.

Architectural Comparison Matrix

Model	Rep	Order	Pooling	Params
LogLin	BoW	No	None	25k
S-MLP	BoW	No	None	2.10M
BiLSTM	Seq	**Yes**	**Tri-Pool**	1.97M
Transf	Seq	**Yes**	**Tri-Pool**	1.38M

Why Deep MLP Failed on BoW

The Deep MLP failed (0.6122 Macro-F1 vs 0.6270 for Shallow MLP) because feedforward depth cannot recover lost word order; it merely amplifies co-adaptation on sparse bags of words.

5. Evaluation Setup & Bayesian ROPE Analysis

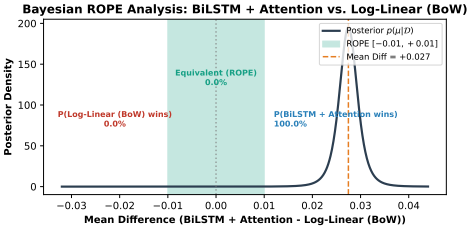
Topic 5: Stratified Cross-Validation & Posterior Probabilities

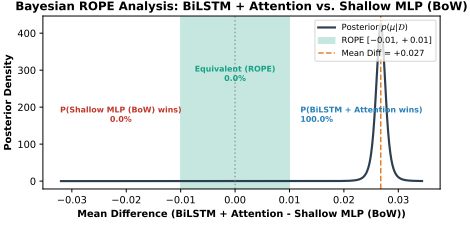
Rigorous Evaluation Protocol:

- **5-Fold Stratified CV** on Train ($N = 45,615$). Zero test leakage.
- **Primary Metric**: Macro-F1 (unweighted class average).
- **Correlated Bayesian t-test** (Benavoli et al., 2017): Corrects for fold dependency ($r = 1/(K-1) = 0.25$).
- **ROPE**: $[-\mathbf{0.01}, +\mathbf{0.01}]$ ($\pm 1\%$ Macro-F1 equivalence band).

Decisive Bayesian Posterior Probabilities:

- **BiLSTM vs. Log-Linear**: $\mathbf{P}(\text{BiLSTM} > \text{LogLin}) = \mathbf{99.96}\%$ (Mean Diff $+\mathbf{0.0275}$, 95% HDI $[+0.022, +0.033]$).
- **BiLSTM vs. Shallow MLP**: $\mathbf{P}(\text{BiLSTM} > \text{S-MLP}) = \mathbf{100.00}\%$ (Mean Diff $+\mathbf{0.0268}$, Effect Size 30.4).
- **BiLSTM vs. Deep MLP**: $\mathbf{P}(\text{BiLSTM} > \text{Deep}) = \mathbf{99.97}\%$ (Mean Diff $+\mathbf{0.0416}$).
- **BoW vs. TF-IDF**: $\mathbf{P}(\text{BoW} > \text{TF-IDF}) = \mathbf{99.43}\%$.





6. Results, Error Analysis & Final Decision

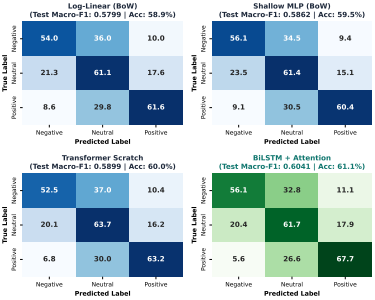
Benchmark

Performance Summary (CV & Official Test Set $N = 12,284$):

Model	Rep	Macro-F1 (CV)	Macro-F1 (Test)
`loglin_bow`	BoW	0.6263 ± 0.005	0.5799 (58.9% Acc)
`shallow_mlp_bow`	BoW	0.6270 ± 0.005	0.5862 (59.5% Acc)
`deep_mlp_bow`	BoW	0.6122 ± 0.006	– (Ablation)
`transformer`	Seq	0.6288 ± 0.005	0.5899 (60.0% Acc)
`bilstm_attn`	**Seq**	$\mathbf{0.6538 \pm 0.005}$	**0.6041** (**61.1**% Acc)

Error Analysis & Oral Defense Takeaways:

- **1,195 Errors Corrected by BiLSTM**: Correctly resolved negation clauses and syntactic modifiers where BoW generated false alarms.
- **Leak-Free Protocol**: 5-fold ensemble with internal fold validation checkpointing; official test evaluated strictly once with unbiased priors.
- **Champion**: **BiLSTM + Attention** (Highest CV Macro-F1 0.6538, decisive Bayesian posterior $> 99.9\%$, highest test generalization).



Final Model Decision for Oral Exam

BiLSTM + Attention is our champion. Combining in-domain PPMI-SVD subwords, multi-pooling, and recurrent sequence modeling overcomes the orderless BoW limit with full statistical confidence.